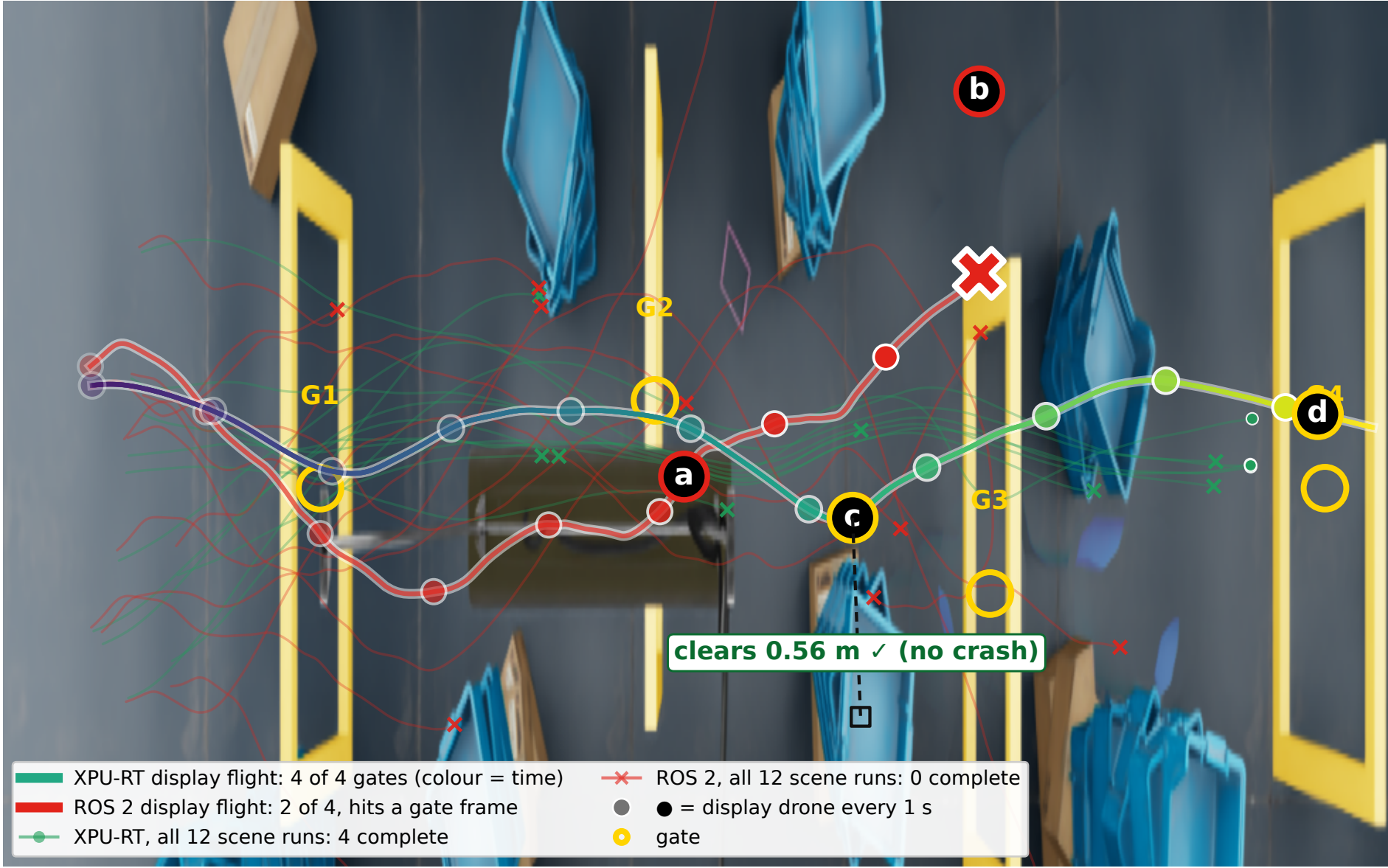


XPU-RT derives automatic what ROS 2 needs an expert to hand-build, and beats ROS 2 as normally deployed — on gates cleared, ahead in 7 of 13 paired censuses, level in 6, behind in 0

With control chained to perception, as ROS 2 is normally written, XPU-RT is ahead in 6 of 9. ROS 2 draws level when hand-pinned (5 censuses) or at a 40 Hz camera; with a hand-added 100 Hz control timer it is level in 3 of 4 (the one behind keeps a 242 ms camera→goal chain). ROS 2 completes more courses in 3 (45 Hz hand-pinned to 6 cores: 8 vs 7; 40 Hz two YOLO instances over all 8 harts: 7 vs 5; 45 Hz hand-pinned + control timer: 19 vs 18), each inside its Newcombe interval. XPU-RT schedules control in its own 100 Hz slot at every camera rate; its solver finds the placement without hand-pinning.

A The display scene, 36 Hz camera, 1.4 m/s: solver-placed XPU-RT (100 Hz control) against ROS 2 over all 8 harts — two YOLO pools and a nav pool, control chained (36 Hz)



B Every paired census: the same cells and replicates per arm

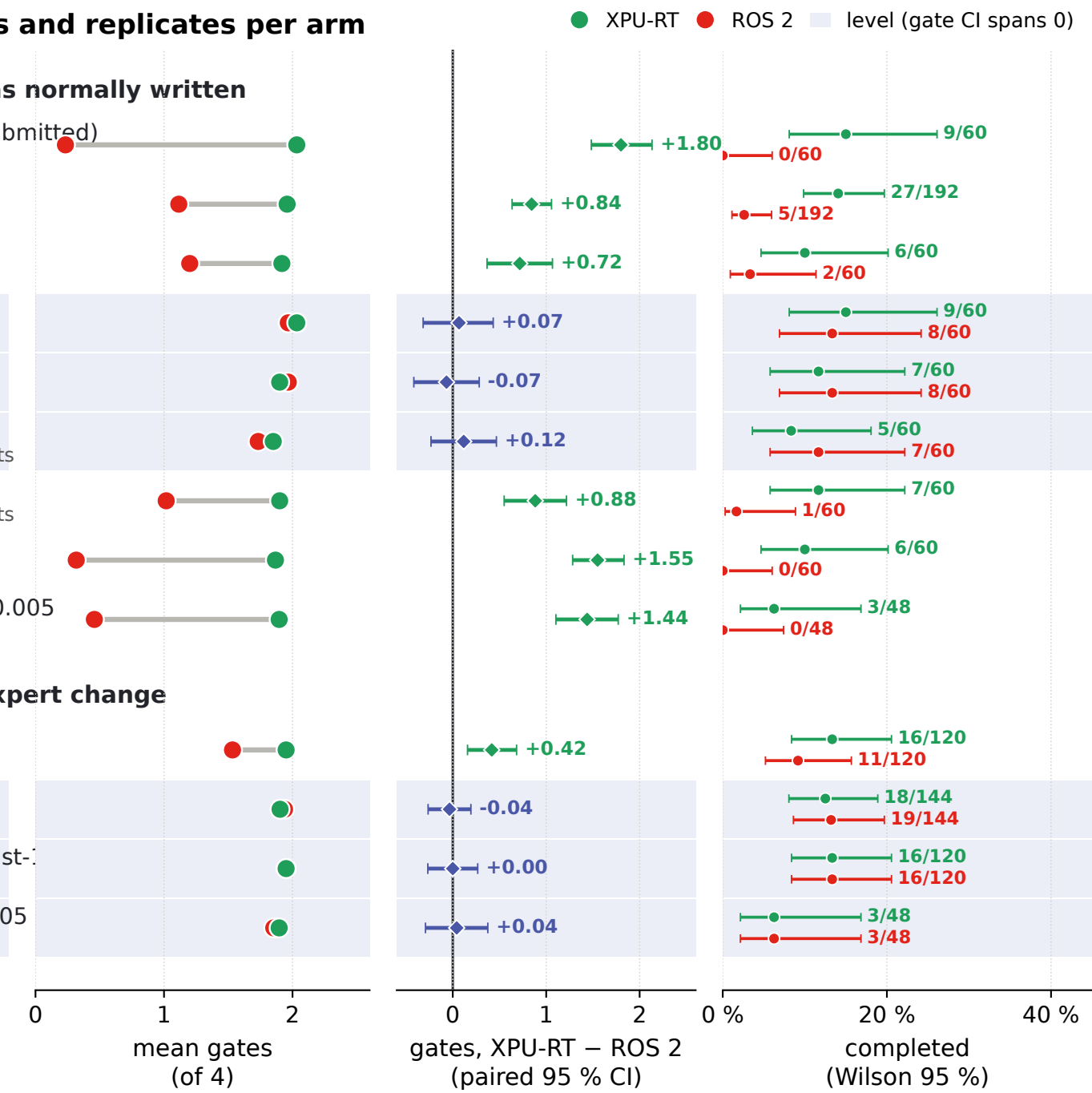
control chained to perception — ROS 2 as normally written

45 Hz · serial YOLO, ctrl in goal callback (as submitted)
ROS 2 20 Hz ctrl · vs XPU-RT CP-SAT
45 Hz · 4-hart YOLO pool, unpinned
ROS 2 38 Hz ctrl · vs XPU-RT CP-SAT
45 Hz · two YOLO instances over all 8 harts
ROS 2 45 Hz ctrl · vs XPU-RT CP-SAT
45 Hz · hand-pinned to 6 cores
ROS 2 38 Hz ctrl · vs XPU-RT CP-SAT
45 Hz · hand-pinned to 6 cores
ROS 2 38 Hz ctrl · vs XPU-RT CP-SAT, solver-placed
40 Hz · two YOLO instances over all 8 harts
ROS 2 40 Hz ctrl · vs XPU-RT greedy on measured costs
36 Hz · two YOLO instances over all 8 harts
ROS 2 36 Hz ctrl · vs XPU-RT greedy on measured costs
30 Hz · two YOLO instances over all 8 harts
ROS 2 30 Hz ctrl · vs XPU-RT CP-SAT
30 Hz · partitioned, nav on the E cores · gain 0.005
ROS 2 30 Hz ctrl · vs XPU-RT CP-SAT, solver-placed

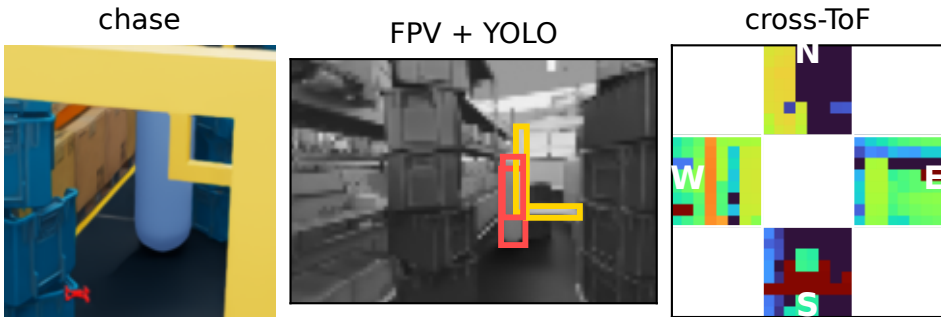
control on its own 100 Hz timer — the expert change

45 Hz · 4-hart YOLO pool + control timer
ROS 2 100 Hz ctrl · vs XPU-RT CP-SAT
45 Hz · hand-pinned + control timer
ROS 2 100 Hz ctrl · vs XPU-RT CP-SAT
45 Hz · hand-pinned + control timer + keep-last-
ROS 2 100 Hz ctrl · vs XPU-RT CP-SAT
30 Hz · hand-pinned + control timer · gain 0.005
ROS 2 100 Hz ctrl · vs XPU-RT CP-SAT, solver-placed

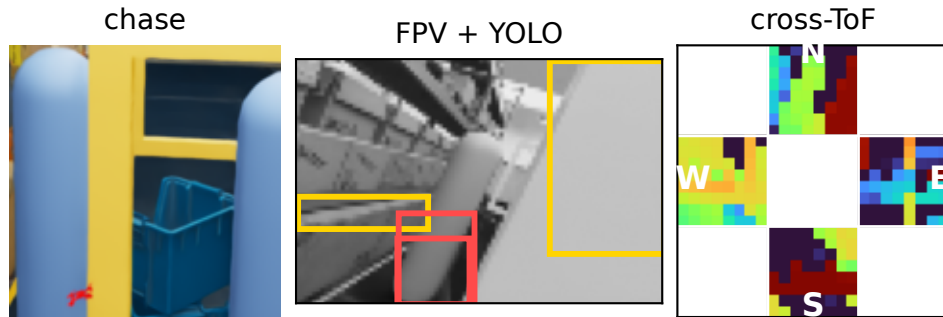
Isaac Sim, K1 timing replayed · 2184 flights



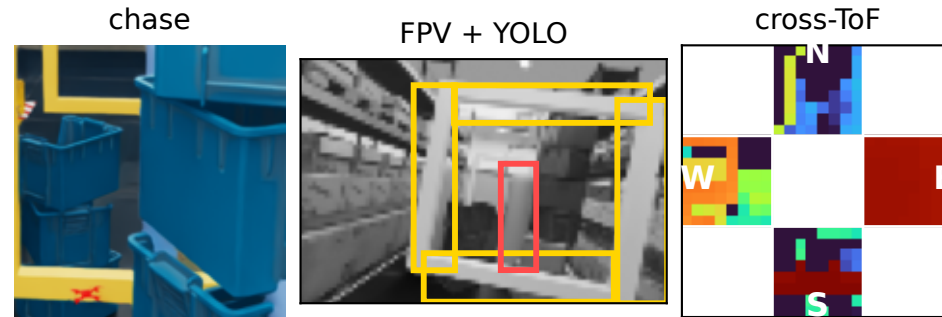
a ROS 2 36 Hz · clears G2 · t=6.1s



b ROS 2 36 Hz · hits gate frame after G2 · t=8.7s



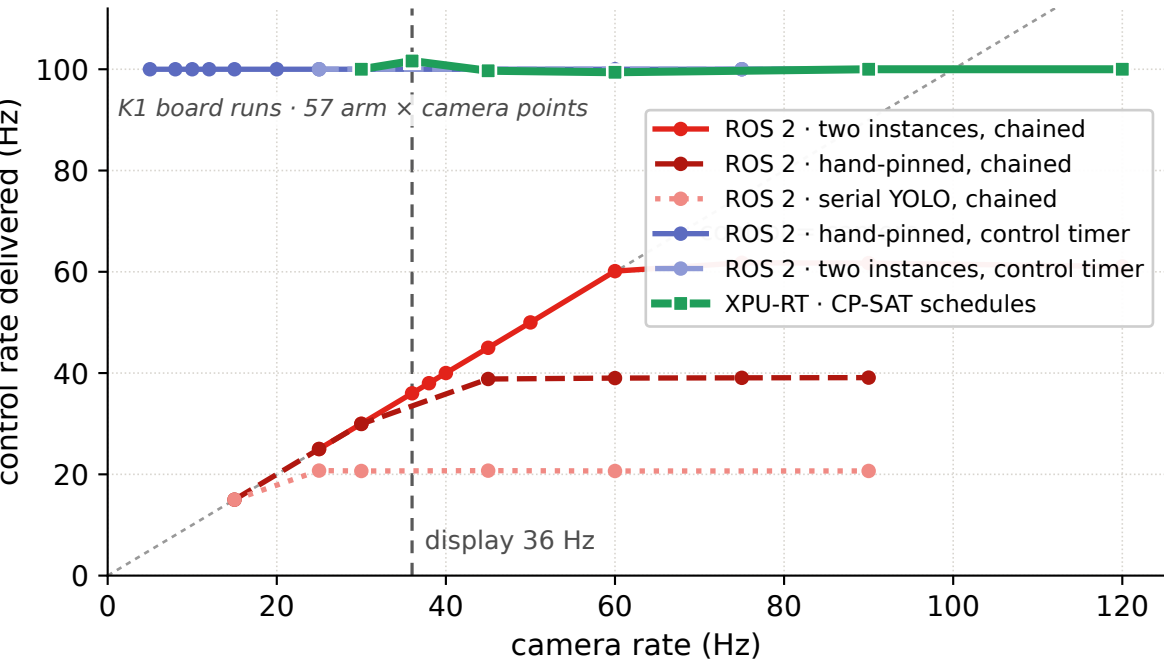
c XPU-RT 100 Hz · clears crate 0.56 m · t=7.2s



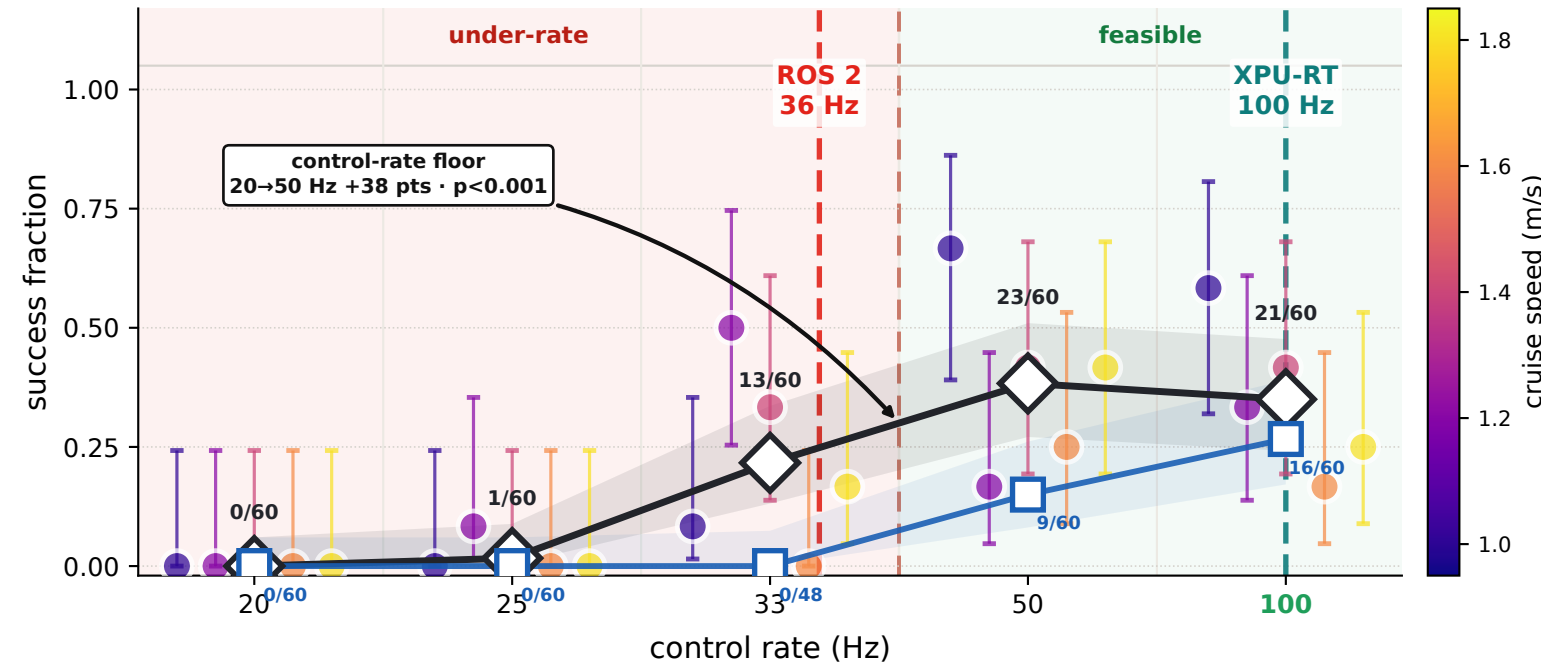
d XPU-RT 100 Hz · reaches gate · t=11.1s



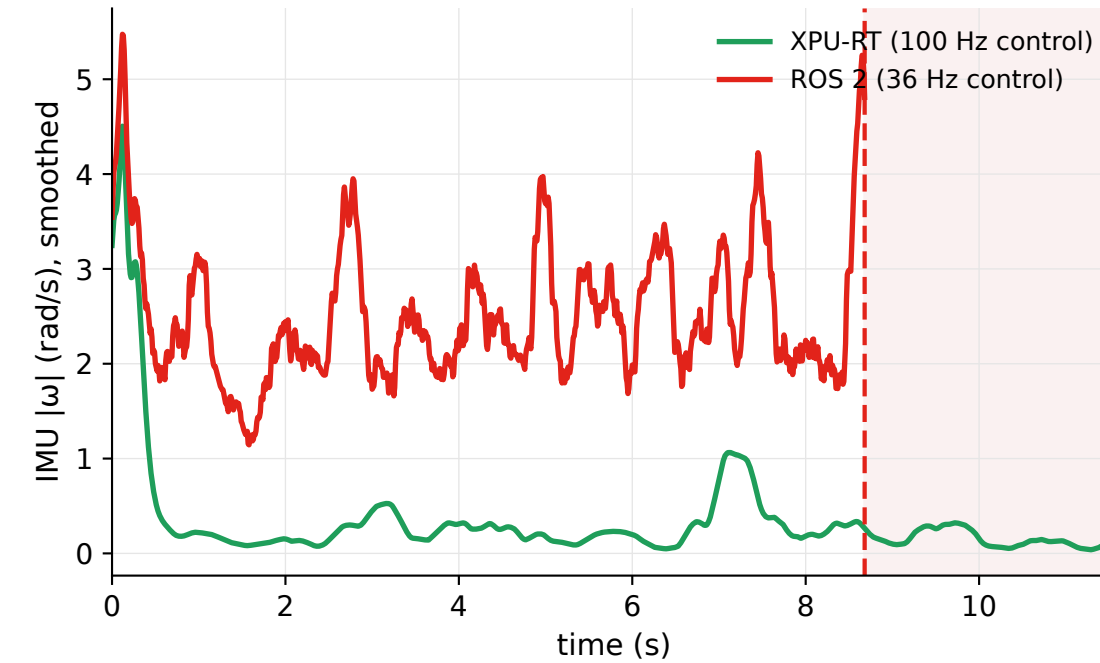
C Mechanism, measured on the K1: chained control tracks the camera; a timer or XPU-RT's slot holds 100 Hz



D Why the rate matters: completion against control rate, rate injected in Isaac Sim (300 flights), the display arms marked



E Inside the display pair: body rate (the baseline thrashes at its control rate)



I Onboard K1 schedule, measured at the 36 Hz camera — XPU-RT · solver-placed CP-SAT places the kernels over all 8 harts: camera→control 26 ms, control every 9.9 ms, every frame on time · YOLO on 8 harts, 6 traced, control on 8 harts, nav on 8 harts ROS 2 (5 processes · single executor · chained control · unpinned): camera→control 37 ms, control every 27.8 ms, 0/352 frames late · YOLO on 8 harts, 6 traced, nav on 5 harts, 4 traced, control on 3 harts

